AI Automation Risk — Workforce Analytics (Power BI)

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A Power BI project scoring AI-automation risk across 3,000 global roles on a star-schema model with custom DAX, finding that occupation — not geography or salary, predicts who is most exposed.

Business problem & objective

Organisations planning for AI disruption need to know who is most exposed, so upskilling budgets target the right people. The common assumptions that high earners, or workers in developed economies, are safest are an untested assumption.

The objective is to build an interactive analytics solution that scores AI-automation risk across roles, industries, countries and salary bands, and tests whether geography or pay actually protects a role.

Data & method

Dataset. 3,000 global employee records. Primary data is a public Kaggle "AI Impact on Jobs" dataset (role, industry, country, salary, experience, AI-replacement-risk score). A secondary complementary table (training hours, digital-skill level, productivity, innovation) was synthetically generated and is used for method demonstration only — not treated as real-world evidence.

Preparation. Cleaned in Excel: missing values imputed, outliers assessed with the interquartile-range (IQR) method.

Model. Star schema in Power BI — a central fact table joined on `Employee_ID` to four dimension tables (Country, Industry, Education, Company Size). This keeps queries fast and slicing clean.

Measures. DAX measures including `CALCULATE`, `DIVIDE`, `RANKX`, `COUNTROWS`, and custom correlation measures (AI risk vs salary, vs future demand, experience vs salary) driving KPIs such as High Risk %, average AI risk, and role risk ranking.

Dashboard. Interactive report with KPI cards, a role-risk ranking, a salary-vs-risk scatter, a geographic map, and slicers for drill-down by industry, country and education.

Key findings

Three evidenced findings from the primary data; 27% of the 3,000 roles fall in the high-risk band.

Finding	Evidence
Occupation predicts risk far more than geography	Risk varied 0.46–0.57 across roles (UX Designer 0.46 vs DevOps Engineer 0.57) but only 0.48–0.53 across countries (Singapore 0.48 vs Pakistan 0.53)
Salary is not a shield	No clear salary–risk relationship; Finance pairs the highest average pay (~$132K) with a high 0.53 risk
Roles reshape rather than vanish	Demand stays broadly steady across experience levels, consistent with AI redefining roles over eliminating them

Recommendation

Prioritise upskilling by *role*, not by salary band or location. Because exposure tracks what a job does rather than where it sits or what it pays, workforce planning should target the highest-risk occupations first. identifying them early and moving those people toward tasks AI complements rather than replaces.

Limitations & integrity

- **Synthetic variables.** Training hours, digital skill, productivity and innovation were AI-generated. Any pattern involving them demonstrates method only and is not claimed as a real-world result.
- **Correlation, not causation.** The findings describe associations in the data, not proven cause and effect.
- **“2030” is framing, not a forecast.** The dataset is cross-sectional, the project contains no predictive time-series model, so “demand stays steady” is a trend observation, not a forecast.
- **Public dataset.** Primary data is an educational Kaggle set, suitable for a skills portfolio rather than live decision-making.

Skills demonstrated

- **Data modelling** — star-schema design, fact/dimension relationships, foreign keys
- **DAX** — CALCULATE, RANKX, DIVIDE, COUNTROWS, custom correlation measures
- **Data preparation** — imputation, IQR outlier handling, multi-source integration
- **Dashboard design** — KPI cards, ranking, scatter, map, interactive slicers
- **Analytical communication** — translating a model into three findings and one recommendation, with stated limitations

Tools: Microsoft Power BI, DAX, Microsoft Excel.